

Geometric Sequences and Series



Geometric sequences

- 1 Consider the sequence 3, 6, 12, 24, ...
 - a State the common ratio.
 - b Write a formula for the n th term.
 - c Find a_{10} .
 - 2 In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.
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Geometric series

- 3 Find the sum of the first 8 terms of $5 + 10 + 20 + \dots$
 - 4 Evaluate each infinite sum, or state that it diverges:
 - a $18 + 6 + 2 + \frac{2}{3} + \dots$
 - b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$
 - c $2 + 3 + 4.5 + \dots$
 - 5 Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.
 - 6 A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.
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Geometric means and mixed sequences

- 7 Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that 4, $b_1, b_2, 108$ is geometric).
 - 8 Determine whether each sequence is arithmetic, geometric, or neither:
 - a 2, 6, 18, 54, ...
 - b 5, 8, 11, 14, ...
 - c 1, 4, 9, 16, ...
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Applications

- 9** A car worth 80,000 riyals depreciates by 15% per year.
- a** Write a formula for its value after n years.
 - b** Find its value after 5 years, to the nearest riyal.
- 10** A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).